HIGH PRESSURE PROCESS WATER PUMPS
2000-F17-PID-PR-10002
HP PROCESS WATER
17-0174-PW-PE16-160
MILL WATER PUMPS
2000-F24-PID-PR-10001
PROCESS WATER
17-0360-PW-PE8-225
17-0361-PW-SS1-200
FLOCCULANT DOSING PUMPS
2000-F72-PID-PR-10004
FLOCCULANT
17-0371-FL-PE8-90
17-0372-FL-MS1-80
FLOTATION AREA SUMP PUMP 2
2000-F16-PID-PR-10003
SPILLAGE
16-1208-FD-PE10-160
FLOTATION AREA SUMP PUMP 3
2000-F16-PID-PR-10004
16-1214-FD-PE10-160
CLEANER AREA SUMP PUMP
2000-F18-PID-PR-10001
18-1216-FD-PE10-110
FLOCCULANT AREA SUMP PUMP
2000-F72-PID-PR-10004
72-1076-FL-PE8-90
FLOTATION TAILINGS THICKENER UNDERFLOW PUMP
2000-F17-PID-PR-10002
FLOTATION TAILINGS THICKENER UNDERFLOW
17-0383-CON-PE10-500
MILL WATER PUMPS
2000-F24-PID-PR-10001
17-0187-PW-PE8-500
500VBFD1L2
F17-10846
FLOTATION CELLS DRAIN PUMP
2000-F16-PID-PR-10003
SCAVENGER TAILS
16-1290-FS-PE10-280
FLOTATION AREA SUMP PUMP 1
2000-F16-PID-PR-10001
16-1206-FD-PE10-160
RAW WATER PUMPS
2000-F24-PID-PR-10002
RAW WATER
17-0300-BW-PE8-63
17-0975-BW-SS3-50
17-1424-BW-SS3-25
FLOTATION DISCHARGE BOX
2000-F16-PID-PR-10003
SCAVENGER TAILS
16-1248-FS-RL1-1200
FIT 17001
FE 17001
FIC 17001
FCV 17001
f(x)
DILUTION FACTOR
DETAIL 1
FIT 17003
FE 17003
PI 17002
PT 17002
FI 17003
17-0982-PW-PE16-160
17-0983-PW-PE16-160
F17-XM-413
FLOTATION TAILINGS THICKENER FLOCCULANT INLINE MIXER
F24-XM-451
F24-XM-452
F24-XM-453
F24-XM-454
F24-XM-455
F24-XM-456
FLOTATION TAILINGS THICKENER WATER MONITORS (6 OFF)
NOTE 7
SP MC-002
SP HS-100
SP MC-001
SP @ BRIDGE OF THICKENER F17-TH-401
MANTA SUB START/STOP
HS 17038
MANTA SUB RUNNING
MANTA SUB FAULT
MANTA SENSOR FAILED TO RETURN HOME ALARM
EMERGENCY STOP
MAX. RAKE LEVEL
MIN. RAKE LEVEL
F17-XM-572
MANTA SUB
F17-HU-406
FLOTATION TAILINGS THICKENER RAKE DRIVE HPU
F17-HU-414
FLOTATION TAILINGS THICKENER RAKE LIFT HPU
VENDOR PACKAGE
2000-ME0015-E02-50001 TO 50006
24VDC POWER HEALTHY
RAKE DRIVE FORWARD SOLENOID RELAY
RAKE DRIVE REVERSE SOLENOID RELAY
RAKE LIFT RAISE SOLENOID RELAY
RAKE LIFT LOWER SOLENOID RELAY
RAKE DRIVE RUNNING FEEDBACK
LOCAL MODE ENABLE FEEDBACK
TORQUE INDICATION
85% TORQUE FROM DCS TO LCP
50% TORQUE FROM DCS TO LCP
40% TORQUE FROM DCS TO LCP
30% TORQUE FROM DCS TO LCP
RAKE LIFT LOCAL RAISE REQUEST
RAKE LIFT LOCAL LOWER REQUEST
REVERSE JOG LOCAL START REQUEST
ALARM ACKNOWLEDGE FROM DCS TO LCP
REVERSE JOG ACKNOWLEDGE FROM DCS TO LCP
RAKE DRIVE FORWARD LOCAL START REQUEST
RAKE DRIVE LOCAL STOP REQUEST
RAKE DRIVE OIL TANK COOLING FAN OUTLET FILTER DIFF PRESSURE (WORKING)
RAKE DRIVE OIL TANK PUMP OUTLET FILTER DIFF PRESSURE
RAKE DRIVE OIL TANK COOLING FAN OUTLET FILTER DIFF PRESSURE (STANDBY)
RAKE DRIVE OIL TANK OUTLET OIL DIFF PRESSURE
RAKE LIFT POSITION
RAKE DRIVE OIL TANK TEMPERATURE HIGH
RAKE DRIVE OIL TANK TEMPERATURE HIGH
RAKE DRIVE OIL TANK LEVEL LOW LOW
RAKE DRIVE OIL TANK LEVEL LOW
RAKE LIFT OIL TANK LEVEL LOW
RAKE LIFT OIL TANK TEMPERATURE HIGH
RAKE LIFT OIL TANK LEVEL LOW LOW
RAKE LIFT OIL TANK TEMPERATURE HIGH HIGH
RAKE LIFT OIL TANK PUMP OUTLET FILTER DIFF PRESSURE
GEAR REDUCER OIL LEVEL LOW
GEAR REDUCER OIL LEVEL LOW
RAKE ZERO SPEED
F17-PO-408 THICKENER RAKE DRIVE PUMP SUCTION VALVE POSITION SWITCH OPENED
FIC17003
2000-F72-PID-PR-10004
SIGNAL
TURBIDITY
LIT 17012
BED LEVEL METER MANTA SUB
RAKE HEIGHT LO
RAKE HEIGHT HI
RAKE UNDERSPEED SWITCH
GEARBOX LEVEL SWITCH LOW
GEARBOX LEVEL SWITCH LOW
RAKE HEIGHT POSITION
F17-VL-458
FLOTATION TAILINGS THICKENER OVERFLOW LAUNDER
F17-BX-464
FLOTATION TAILINGS THICKENER OVERFLOW BOX
F17-XM-572
FLOTATION TAILINGS THICKENER MANTA SUB
F17-VL-490
FLOTATION TAILINGS THICKENER OVERFLOW PIPE LAUNDER 1
F17-VL-491
FLOTATION TAILINGS THICKENER OVERFLOW PIPE LAUNDER 2
MECH
PROCESS WATER TANK
2000-F24-PID-PR-10001
FLOTATION TAILINGS THICKENER OVERFLOW
FIC17121
2000-F17-PID-PR-10002
SIGNAL
BED PRESSURE
PIC 17011
BED MASS METER
PE 17011
CCTV
F17-CAM-001
F17-XM-572
FLOTATION TAILINGS THICKENER MANTA SUB
SP MC-001
SP HS-025
F17-SA-416
FLOTATION TAILINGS THICKENER FEED SAMPLER
VENDOR PACKAGE
2000-ME074-J02-10007
F17-BX-410
FLOTATION TAILINGS THICKENER RECIRCULATION FEED BOX
F17-HU-422
FLOTATION TAILINGS THICKENER FEED BOX VALVE HYDRAULIC POWER UNIT
F17-PP-459
THICKENER FEED STANDPIPE
FLOTATION TAILINGS P-DUC PUMP
NOTE 6
F17-TH-401
FLOTATION TAILINGS THICKENER
F17-XM-474
F17-XM-471
F17-XM-613
F17-XM-472
F17-XM-614
F17-XM-473
FLOTATION TAILINGS THICKENER OVERFLOW LAUNDER DART VALVES
(6 OFF)
NOTE 5
VSD
1200VKGD1L2
FL
DN1200
DN50
TO GRADE
17-0497-FS-RL1-1200
17-1742-FS-RL1-1200
17-1743-FS-RL1-80
80VKGD1L1
F17-10851
LIC 17004
LT 17004
LI 17004
DETAIL 3
HS 17005
XV 17005
80VKGD1L1
FL
F16-HP-456
FLOTATION TAILS SAMPLE HOPPER
DETAIL 4
HS 17006
XV 17006
16-1515-FS-SS3-50
TO GRADE
RL1
PE10
90x63
16-0660-FS-RL1-80
16-2140-FS-RL1-80
80x65
SP HS-074
F16-PP-634
FINAL TAILS SAMPLE PUMP
BURST
XA 17008
XS 17008
DETAIL 4
HS 17007
XV 17007
16-1448-FS-SS3-50
TO GRADE
ZAH 17144
PAH 17143
TI 17141
PI 17140
LI 17142
LAH 17081
LSH 17081
F17-PS-427
FLOTATION TAILINGS THICKENER AREA SUMP PUMP 1
NOTE 7
17-0368-FD-PE10-225
17-0367-FD-RL1-200
16-0496-FS-PE10-63
16-3072-FS-PE10-110
TO GRADE
17-0498-CON-RL1-300
300VKGD1L1
F17-10863
TO GRADE
17-0364-CON-RL1-600
FLOTATION TAILINGS THICKENER UNDERFLOW PUMP 1
2000-F17-PID-PR-10002
FLOTATION TAILINGS THICKENER UNDERFLOW
17-0374-CON-RL1-600
FLOTATION TAILINGS THICKENER UNDERFLOW PUMP 1
2000-F17-PID-PR-10002
FLOTATION TAILINGS THICKENER UNDERFLOW
17-0365-CON-RL1-600
FLOTATION TAILINGS THICKENER UNDERFLOW PUMP 2
2000-F17-PID-PR-10002
FLOTATION TAILINGS THICKENER UNDERFLOW
17-0377-CON-RL1-600
FLOTATION TAILINGS THICKENER UNDERFLOW PUMP 2
2000-F17-PID-PR-10002
FLOTATION TAILINGS THICKENER UNDERFLOW
1:100
16-0661-FS-PE10-63
FLOTATION PARTICLE SIZE ANALYSER
2000-F16-PID-PR-10005
FINAL TAILS SAMPLE

NOTES:

1. REFER TO CONTROL PHILOSOPHY FOR COMPREHENSIVE LIST OF PROCESS INTERLOCKS AND CONTROL FUNCTIONS.
2. STAGE 2.
3. TIE IN POINT LOCATED ON F17-BX-410.
4. SIGNAL FROM FLOTATION THICKENER LCP TO MCC/DCS.
5. FLOCCULANT IS BEING DISTRIBUTED WITHIN VENDOR PACKAGE.
6. THERE ARE 3 OFF SPARE NOZZLES (DN100, DN200 & DN60 AVAILABLE).
7. FLANGED ASME B16.5 CLASS 150.

DRG No.	REFERENCES
2000-F00-PID-PR-10007	LEGEND SHEET 7 - PIPING AND INSTRUMENTATION DIAGRAM
2000-F00-PID-PR-10006	LEGEND SHEET 6 - PIPING AND INSTRUMENTATION DIAGRAM
2000-F00-PID-PR-10005	LEGEND SHEET 5 - PIPING AND INSTRUMENTATION DIAGRAM
2000-F00-PID-PR-10004	LEGEND SHEET 4 - PIPING AND INSTRUMENTATION DIAGRAM
2000-F00-PID-PR-10003	LEGEND SHEET 3 - PIPING AND INSTRUMENTATION DIAGRAM
2000-F00-PID-PR-10002	LEGEND SHEET 2 - PIPING AND INSTRUMENTATION DIAGRAM
2000-F00-PID-PR-10001	LEGEND SHEET 1 - PIPING AND INSTRUMENTATION DIAGRAM

REV	DATE	DESCRIPTION	DRAW	CHECK	DESIGN	TECH APP	PROJ APP
			AH		JBe		
2	12SEP25	LAUNDERS AND GENERAL REVISIONS ADDED	AH	MM	JBe	DA	DM
1	19NOV24	RE-ISSUED FOR CONSTRUCTION	NB	JB	JB	LS	DM
0	06SEP24	APPROVED FOR CONSTRUCTION	AK	JB	JB	LS	DM

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PRIMERO PROJECT NUMBER: 23517

PROJECT: NORTHERN STAR RESOURCES LTD
KCGM GROWTH PROJECT

TITLE: FLOTATION TAILINGS PRE-LEACH THICKENING
FLOTATION TAILINGS THICKENING - SHEET 1 - STAGE 1
PIPING AND INSTRUMENTATION DIAGRAM

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SHEET	SCALE	DRG No.	REVISION
A1	NTS	2000-F17-PID-PR-10001	2

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